EventOwner An Exact Marginal Pilot of Audio Visual Source Ownership

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Two audio-visual pairs can contain the same global event inventory while assigning sounds to different visible sources. We construct a small procedural test that makes this relationship necessary for classification. Two event schedules drive rendered rotating objects and four-channel waveforms. Crossing video and audio ownership assignments gives four pairs per scene; every exact video and waveform occurs under both labels. A frozen confirmation study contains 120 scenes and 480 dependent pairs in held-out procedural families. A frequency-resolved joint diagnostic obtains balanced accuracy 0.806, compared with 0.771 for a broadband joint diagnostic, 0.500 for temporal synchrony, and 0.510 for pooled spatial coverage. Scene-cluster intervals preserve the repeated-pair structure. Both joint methods have identical paired-ranking accuracy of 0.996, and performance weakens under overlapping activity and near-colocation. An exploratory 72-window real-recording comparison is inconclusive and favors a simpler spatial reference. The result supports controlled joint construction and evaluation feasibility. It does not establish realistic acoustics, learned source understanding, or real transfer.

Additional Key Words and Phrases: audio visual generation, source ownership, exact marginal controls, procedural data, multimodal evaluation

1 Introduction and Background

An audio-visual generator can align overall timing and sound coverage while attaching an event's sound to the wrong visible object. That error is relational: the video and audio can each look plausible in isolation. A useful diagnostic should distinguish the association between modalities without exploiting label-specific artifacts in either modality.

Existing work already evaluates important parts of this problem. SAVGBench measures spatial audio-video alignment [5]. PAVAS uses physical parameters for video-to-audio synthesis and introduces a physical-attribute consistency evaluation [3]. FlatSounds uses controlled changes in physical factors to audit generation [2]. StereoBind links visual motion tracks and stereo generation [1]. Xu et al. show that shared latent variables alone need not prevent visual shortcuts and compare intervention objectives with supervised counterfactual augmentation [6]. These precedents rule out a claim that causal audio-visual testing or source binding is new.

Our narrower contribution is a source-ownership crossing with exactly matched unimodal marginals, a frozen public implementation, and transparent comparisons between simple joint and non-joint diagnostics. The construction uses actual pixels and waveforms, rather than a predictor supplied with the event state that generated them. It also preserves cases where frequency-resolved scoring fails and cases where its apparent advantage depends on the chosen threshold. The associated exploratory real-data check reports a negative boundary rather than treating procedural success as transfer.

The code, complete numeric results, procedural examples, and paper source ZIP are available at https://github.com/Saurav-K-Aryal/posesplit-pilot. The repository URL is retained for continuity with an earlier, separately archived PoseSplit study of audio-plus-known-pose viewport prediction. That historical study did not use image features in learned prediction and is not evidence for the present joint task.

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2 Methods

2.1 Shared event state and ownership crossings

Each procedural scene contains two fan-like rotating objects, a calibrated common camera/microphone rig, and two independently sampled mechanical-drive envelopes. The envelopes control blade rotation speed in RGB frames and a phenomenological combination of broadband and tonal audio. Source azimuth determines first-order Ambisonics (FOA) directivity; simple delayed reflections and sensor noise provide bounded nuisance variation. The model is not a computational fluid-dynamics or material-acoustic solver and has not been validated as realistic fan acoustics.

Each clip lasts four seconds, with 96 RGB frames at 24 frames/s and 16-kHz four-channel audio. The fixed image size is 128×72 pixels. The spatial methods share a 130-degree camera field of view and W,Y,Z,X FOA convention, with camera-left positive azimuth. Source origins are static. Arbitrary unknown camera/microphone extrinsics are outside this common-rig experiment.

Let V_0, V_1 be the two video assignments of the event schedules to the visible sources, and A_0, A_1 the corresponding audio assignments. Each scene contributes the Cartesian product

$$\{(V_0, A_0), (V_0, A_1), (V_1, A_0), (V_1, A_1)\}, \qquad y(V_i, A_j) = \mathbf{1}[i = j].$$

Each exact video and audio item occurs once with each label. Therefore any deterministic classifier receiving only one modality must obtain balanced accuracy exactly 0.5 on the complete construction, irrespective of capacity. This is a property of the balanced inputs, not a trained unimodal model result. A hash audit verifies the marginal equality scene by scene. Gain, duration, format, and sensor noise are not processed differently by label.

2.2 Development and frozen confirmation

Sixty development scenes use three- and four-blade families. Before confirmation, code, protocol, scalar thresholds, and development results were hashed and frozen. The confirmation run renders 120 scenes in five-blade, six-blade, and asymmetric-blade/layout families. Four stress strata repeat evenly: separated sources, overlapping activation, near-colocation, and partial visual occlusion. Each stratum contains 30 confirmation scenes and 120 pairs. Held-out families are procedural variants, not real-world object classes.

An earlier development attempt used a systematic leading-source order. Its broadband shuffled-scene balanced accuracy of 0.625 exposed an ordering shortcut. Absolute event times and leading-source order were randomized before confirmation, and the original development source and results are retained separately. No confirmation was run for that earlier attempt. All confirmation strata remain in the primary analysis; no difficult cases were removed after scoring.

2.3 Diagnostics and information access

All predictors receive decoded RGB arrays and persisted waveform samples. Source IDs, bounding boxes, physical directions, drive envelopes, ownership labels, and random seeds are used for rendering or grading only.

Frame differences produce a time-resolved optical activity map. A fixed camera projection distributes the activity into 24 azimuth bins from −65 to 65 degrees. Audio uses a 512-sample Hann-window transform with hop 128 and the 250–4200-Hz band. FOA active intensity estimates time-frequency direction; a reliability-weighted kernel distributes energy into the same observed direction bins. A fixed temporal floor subtraction and three-frame smoothing are applied uniformly.

Table 1. Frozen confirmation on 120 scenes and 480 dependent pairs. Confidence intervals resample scenes. The unimodal row follows from exact marginal balancing and is not a trained model measurement.

Diagnostic	Balanced accuracy	95% interval	Paired ranking
Temporal synchrony	0.5000	[0.5000, 0.5000]	0.4917
Pooled spatial	0.5104	[0.5000, 0.5208]	0.5583
Broadband joint	0.7708	[0.7250, 0.8167]	0.9958
Frequency resolved	0.8063	[0.7646, 0.8479]	0.9958
Either modality alone	0.5000	Exact construction	—

The temporal reference correlates activity summed over direction. The pooled-spatial reference correlates direction maps summed over time. The broadband joint diagnostic sums acoustic intensity across frequency before estimating direction, then matches its time-direction map to optical activity. The frequency-resolved diagnostic retains time-frequency directional structure before pooling into that map. These are analytical diagnostics built on ordinary motion and acoustic-intensity processing, not new learned architectures.

Each diagnostic uses one scalar decision threshold chosen on development balanced accuracy. Ties use the median maximizing cut. Feature definitions and thresholds are fixed for confirmation. A separate within-family cross-scene audio shuffle breaks the observed relation while retaining the video. This is a supplementary check; the exact within-scene ownership crossing provides the stronger unimodal shortcut control.

2.4 Analysis and exploratory real component

Balanced accuracy is the mean of sensitivity and specificity. All four pairs from each scene remain together in 2,000 fixed-seed bootstrap samples. We report percentile intervals, paired frequency-resolved-minus-broadband intervals, stress-stratum results, and threshold-free paired ranking. Ranking compares each video's coherent and ownership-swapped audio scores; ties receive half credit. The 480 pairs are not treated as 480 independent scenes.

The real-data component uses STARSS23 [4]. It includes the first eight nonoverlapping four-second windows from each of nine already cached official development-training recordings, for 72 windows. This room/recording cohort had previously been exposed by PoseSplit. The manifest and exact source hashes were fixed before scoring. RGB motion and FOA intensity features are decoded without annotation directions, boxes, or source labels. Each matched window is compared with audio from another recording at the same offset. Recording/room clusters, rather than individual pairs, determine uncertainty. Such real shuffling changes semantics and activity inventory as well as correspondence, so it is not a real-data equivalent of the exact ownership intervention.

3 Results

3.1 Frozen procedural confirmation

Table 1 reports the all-scene confirmation result. Frequency-resolved balanced accuracy is 0.80625, versus 0.77083 for the strong broadband diagnostic. The paired scene-bootstrap difference interval is [0.00625, 0.06458]. Both joint methods reach paired-ranking accuracy 0.99583. Thus the fixed-threshold difference is compatible with a calibration difference; it does not establish generally stronger ranking or an architectural advance.

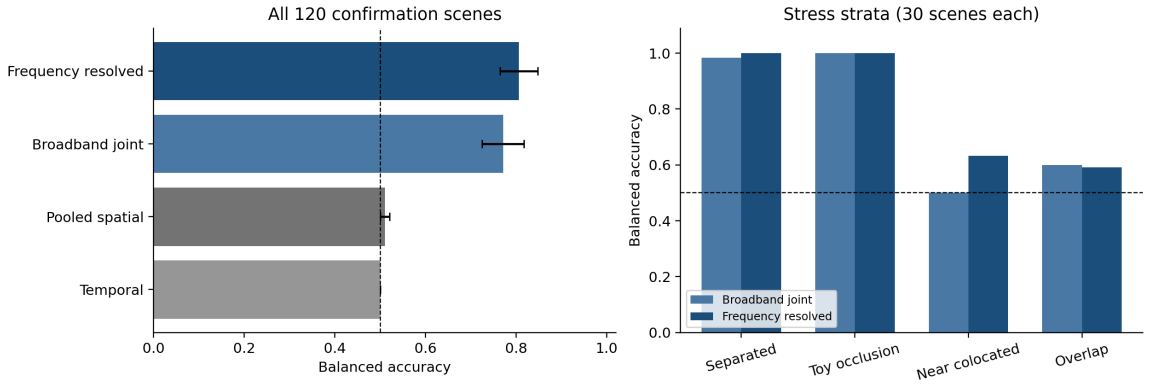


Fig. 1. Confirmation balanced accuracy and stress strata. Error bars at left are scene-bootstrap percentile intervals. All difficult strata remain in the primary result.

Cross-scene shuffled balanced accuracy falls to 0.5000 for frequency-resolved scoring and 0.50625 for broadband scoring. Temporal and pooled-spatial scores remain near chance. Together with the exact marginal audit, these results support a joint relationship test rather than label recovery from either modality alone.

Performance is uneven across strata (Figure 1). Frequency-resolved balanced accuracy reaches 1.000 for separated sources and for the stylized partial occluder, but only 0.6333 for near-colocation and 0.5917 for overlapping activity. Broadband scores are 0.9833, 1.000, 0.5000, and 0.6000 respectively. The frequency-resolved method is therefore slightly worse under overlapping activity. The toy occluder is a limited image construction, not a real occlusion benchmark.

The original confirmation took approximately 51 seconds and peaked near 196 MiB process resident memory on a 128-GB host. No GPU, foundation-model fitting, paid API, or cloud service was used. This is not a direct timing or memory certification on a 48-GB laptop. A separate numerical replay checked every score-derived metric, exact marginal, and bootstrap interval. An independent review regenerated eight randomly selected scenes and reproduced the stored pixels, PCM samples, and scores. The compact public release includes two of those hard-stratum scenes; all other procedural media can be regenerated from stored state.

3.2 Exploratory real-recording result

The real result is less favorable (Figure 2). Frequency-resolved matched-versus-shuffled ranking is 0.6250 with a nine-room interval [0.4861, 0.7500], which includes chance. Broadband ranking is 0.6528 [0.5000, 0.8056]. The simpler pooled-spatial reference reaches 0.7778 [0.6111, 0.9306]; temporal synchrony reaches 0.5556 [0.4583, 0.6528]. This does not establish real source ownership or transfer. It also cautions against selecting a metric only from favorable procedural behavior.

4 Future Work

The next step is a small synthesis study that tests whether explicit source interventions improve a compact residual sound-texture generator beyond ordinary counterfactual augmentation. A coupled model would receive coarse source stems, controls, and scene/video context and emit two residual stems. At fixed physical source identities, changing source A's drive or onset would hold source B's state, timbre, room geometry, and per-source/shared noise fixed. One term

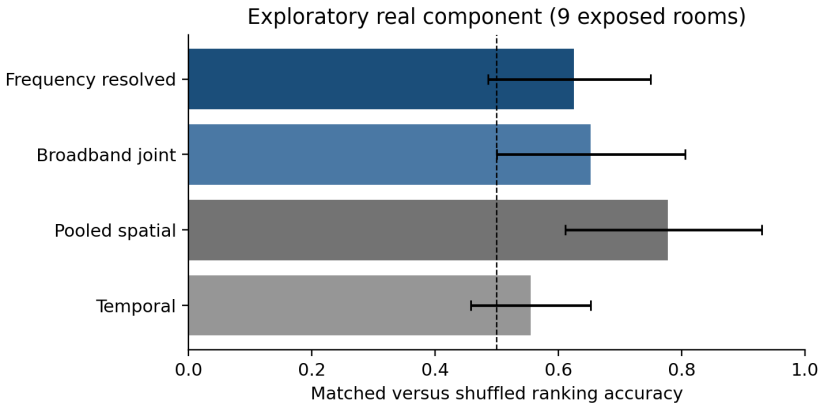


Fig. 2. Exploratory matched-versus-shuffled ranking on 72 windows from nine previously exposed STARSS23 rooms. Error bars resample recording/room clusters. The frequency-resolved interval includes chance and pooled spatial scoring performs better.

would penalize unintended changes in B; another would require A's envelope/onset to respond to the changed control. This is not a source-row permutation objective, which a shared per-source architecture can satisfy by construction.

The main comparator must have the identical coupled architecture, conditioning, counterfactual examples, noise coupling, optimization, and compute, omitting only the added intervention term. A hard independent-source residual model is also required: it prevents cross-source contamination structurally and may be the preferable solution. A small linear counterexample confirms that the coupled-model intervention loss can be nonzero and have a nonzero gradient; it is not evidence of improved learned generation. Reconstruction and texture objectives are still necessary to prevent unresponsive constant outputs. The invariance assumption applies only to physically independent sources under linear room acoustics, not genuine interactions.

A bounded implementation can cap training at 25 million parameters and compare all arms on common state packets and output durations. Permitted texture recordings need per-file source and uploader splits, and upstream terms must remain attached to exports. Physical fidelity, perceptual quality, and source-labeled real transfer each require separate validation. If counterfactual augmentation or hard factorization matches the added term, the simpler generator should be released as the supported result. The current pilot offers a controlled construction and a failure-aware measurement interface for that comparison, rather than evidence that the learned extension will succeed.

5 Conclusion

Exact source-ownership crossings make joint audio-visual information necessary while balancing each modality across labels. In a frozen 120-scene procedural study, simple joint diagnostics outperform timing-only and pooled-spatial references, with a modest fixed-threshold difference between the two joint methods and identical pair ranking. Overlap and near-colocation remain difficult, and the real-recording component is inconclusive. These findings support reproducible joint construction and measurement feasibility and define specific limits for a later source-preserving synthesis study.

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